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Nốt

One positive note every day

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Phùng Thế Quân · SE193519

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Phan Hữu Phước · SE161951

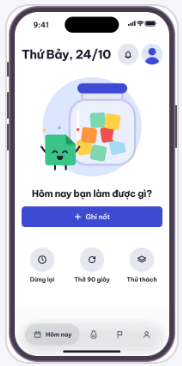


Introduction

Tiny Wins (Nốt) is a mobile app for burned-out final-year students like Minh (22): every evening they drop one small win into a jar instead of staring at an unfinished to-do list. After a rough day, the jar hands back an old win to remember. Small bravery and movement challenges, a one-tap self-control log and a guided journal help them rebuild confidence, one note at a time. Success criterion: log a mood and one win in under 30 seconds.

Overview

From the Lab 2 design to a cloud-connected Flutter app



S02 Hôm nay
↓
/today
↓
TodayScreen
↓
TodayViewModel

→

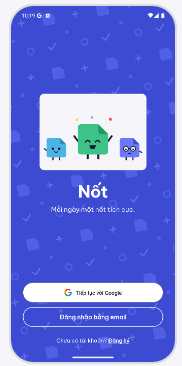
View
TodayScreen

ViewModel · Riverpod
TodayViewModel

Service · interface
NoteRepository

Firebase · lib/firebase
FirestoreNoteRepository

→



Auth
Firestore
Storage

1 Lab 2 design → code

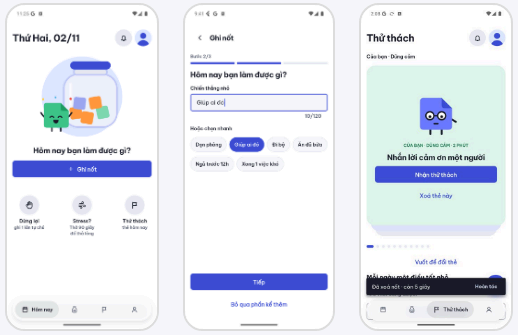
Each Figma screen mapped to a route, Screen and ViewModel

2 MVVM

Screens never touch the SDK; only lib/firebase does

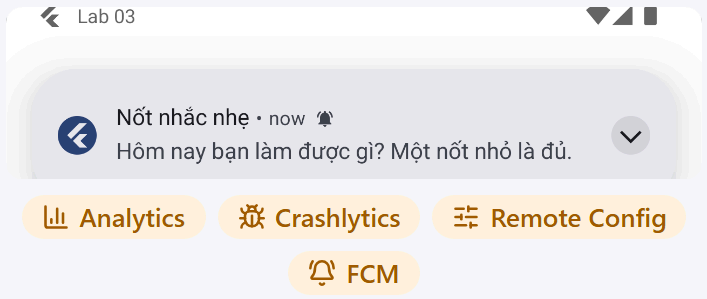
3 Firebase, round 1

Sign-in, notes in Firestore, weekly PDF to Storage

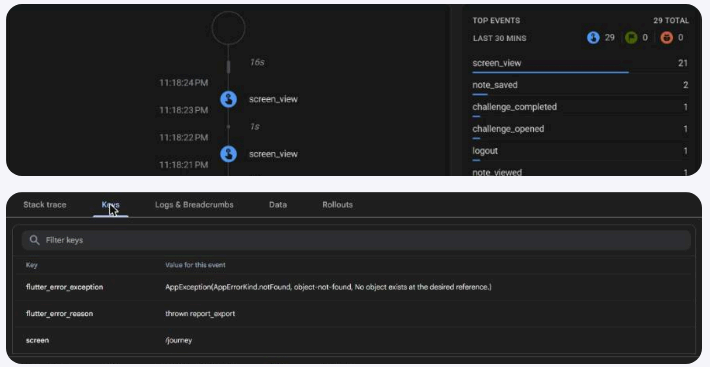


S02 S03 S07

→



→



4 Features

All 12 screens; flows F1 to F3 with their error states

5 Firebase, round 2

Events, crashes, feature flags and push messages

6 Evidence

Console, DebugView, Crashlytics, ARTEMIS · evidence/

Design system in code

Lab 2 tokens become one ThemeData; each Lab 2 component is one reusable widget

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Tokens → ThemeData

Colour → `ColorScheme` app_colors.dart

primary	primary-container	surface	on-surface	error	success	heat/3	heat/5
#3D4FD6	#DDE1FF	#F6F7FB	#1B1D2A	#B3261E	#1F9D63	#FFD24A	#F0524B

Type → `TextTheme` app_typography.dart · Be Vietnam Pro

Nốt nhắc bạn

Nốt nhắc bạn

Nốt nhắc bạn

Nốt nhắc bạn

Nốt nhắc bạn

Display 44 · `displayMedium`

Headline 28 · `headlineMedium`

Title 18 · `titleMedium`

Body 16 · `bodyLarge`

Label 14 · `labelLarge`

Space → `AppSpace`

4 8 12 16 24

Radius → `AppRadius`

8 14 note 16 card full

`AppTheme.light()` wires all of it into one `ThemeData`; screens read the theme, never a raw colour or size

Lab 2 components → widgets/

Button · 5 variants × 4 states	→	`AppButton(variant:, loading:)`
Text field · error, disabled	→	`AppTextField(errorText:)`
Card/Nốt · 4 note types	→	`NoteCard(type:)`
BottomNav · 4 tabs	→	`AppBottomNav(currentIndex:)`
Dialog · confirm, destructive	→	`showAppDialog(kind:)`
Loading · Empty · Error	→	`LoadingState · EmptyState · ErrorState`
MoodOption · selected	→	`MoodOption(mood:, selected:)`
Mascot · 8 states (below)	→	`Mascot(state:)`
Illustration/Hũ chiến thắng	→	`JarIllustration(notes:)`



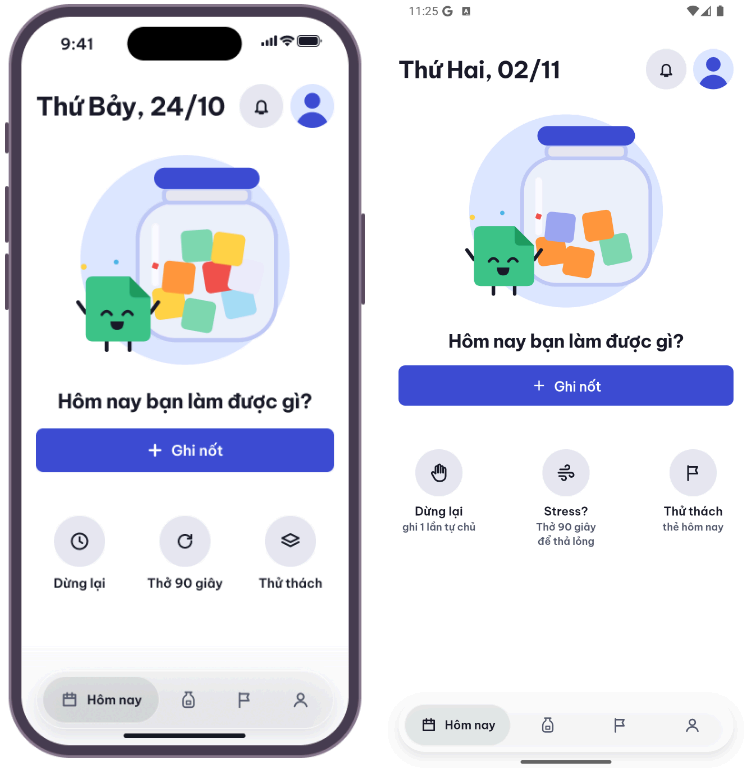
Tụi Nốt: one face per mood and app state, so mood is never colour alone

docs/design-traceability.md §3 components, §4 tokens · lib/theme/ · lib/widgets/

Figma vs app

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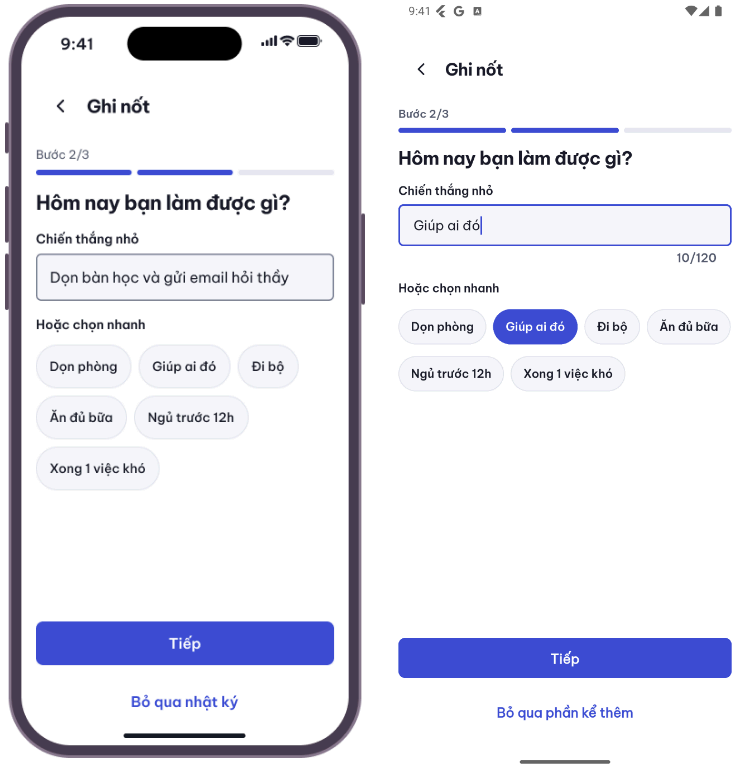
Today



Figma App

- Android status bar (D-04)
- New shortcut icons and captions (D-25)
- "Ghi nốt" asks win or journal first (D-29)

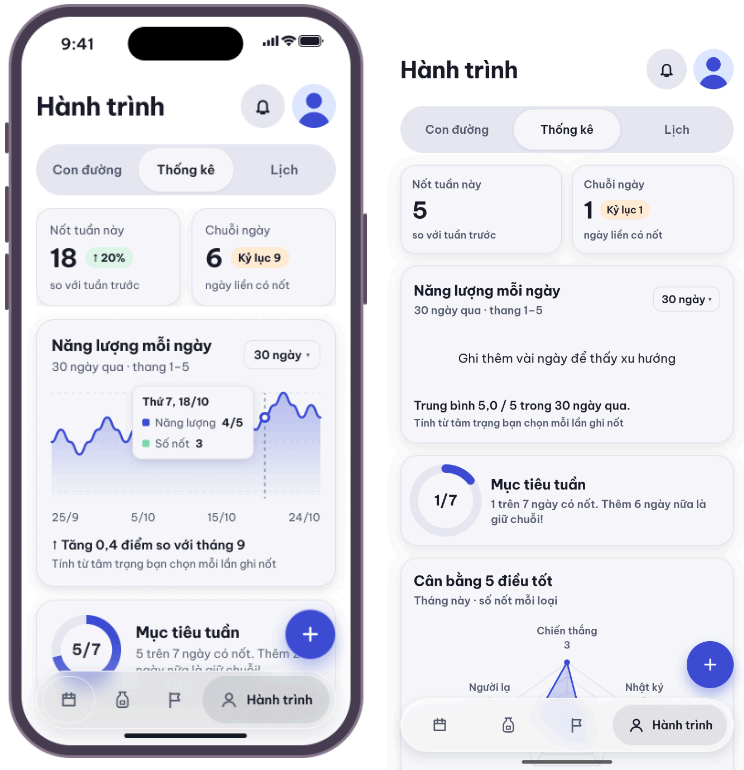
Write a note



Figma App

- "Bỏ qua nhật ký" is now "Bỏ qua phần kể thêm" (D-28)
- 0/120 counter under the field (D-02)
- AI free-tell is planned, not built

Stats



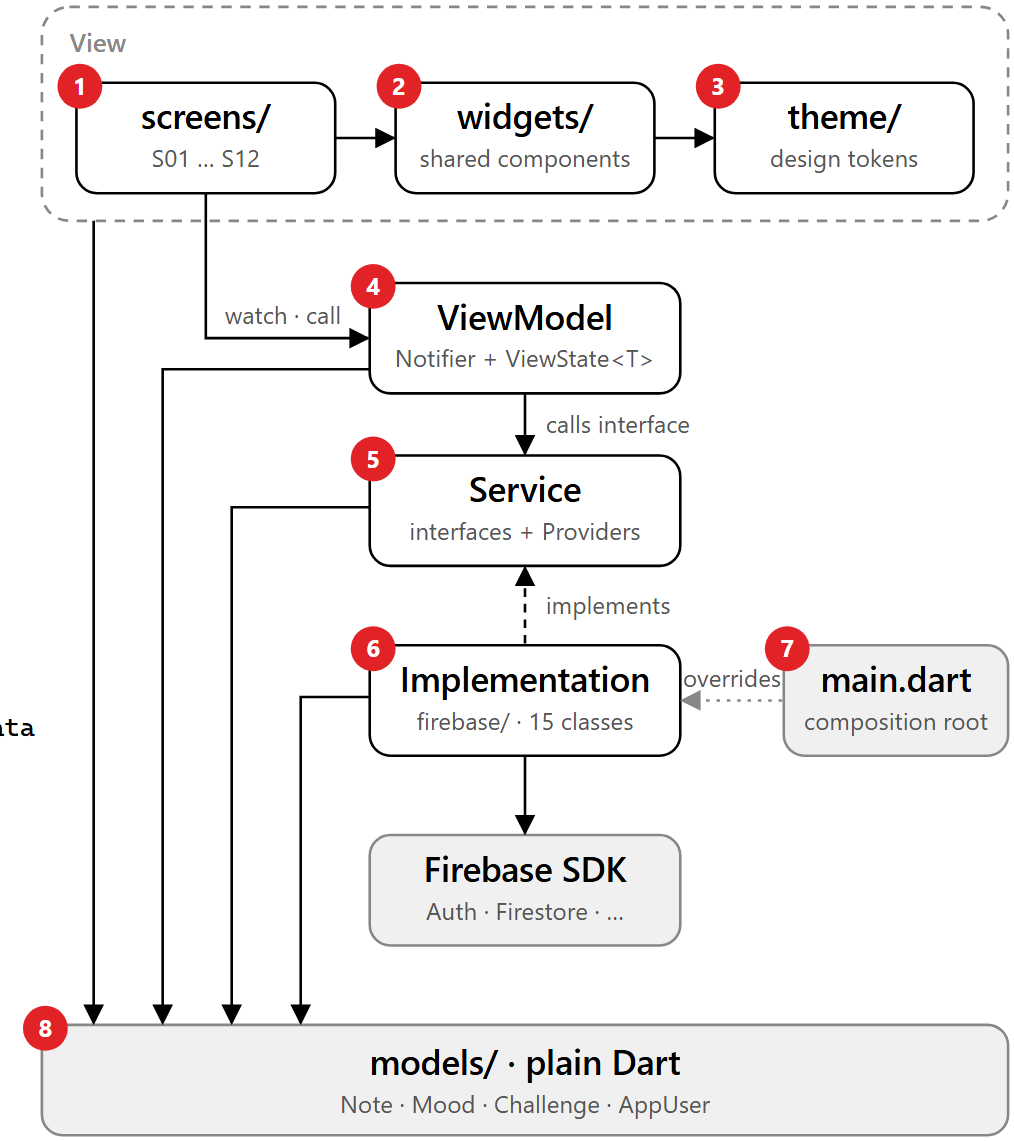
Figma App

- Compared with the previous 30 days (D-18)
- Chart scales to the card width (D-09)
- Weekly goal opens the weekly card (D-06)

docs/design-traceability.md · §1 screens, §5 every deviation with its reason

MVVM

```
lib/
├── main.dart                  composition root
├── app.dart                   MaterialApp.router
├── router/
│   └── auth_redirect.dart     signed out → /welcome
├── theme/
│   └── app_tokens.dart        spacing, radius
├── models/
│   └── note.dart              plain Dart
├── services/
│   └── note_repository.dart
│                              interface + Provider
├── firebase/
│   └── firestore_note_repository.dart
│                              implementation
├── viewmodels/
│   ├── view_state.dart        Loading | Empty | Failure | Data
│   └── today_view_model.dart
│                              Notifier + ViewState
├── screens/today/
│   └── today_screen.dart      switch on ViewState
├── widgets/
│   └── app_button.dart        shared component
└── utils/
    └── dates.dart             pure helpers
```



MVVM in the code: View → ViewModel

S02 Today on GitHub (main). Numbers match the MVVM diagram

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1 screens/ lib/screens/today/today_screen.dart

```
28      @override
29      Widget build(BuildContext context, WidgetRef ref) {
30        final theme = Theme.of(context);
31        final state = ref.watch(todayViewModelProvider);
32        final hasDraft = ref.watch(hasDraftProvider).value ?? false;
33        final summary = switch (state) {
34          Data<TodaySummary>(:final value) => value,
35          _ => null,
```

2 widgets/ lib/widgets/app_button.dart

```
10    /// chạm chạm. Mỗi màn chỉ nên có một nút [AppButtonVariant.filled].
11    class AppButton extends StatelessWidget {
12      const AppButton({
13        required this.label,
14        required this.onPressed,
15        this.variant = AppButtonVariant.filled,
```

3 theme/ lib/theme/app_tokens.dart

```
3   /// Figma collection **Size** (`space/*`) và quy ước lề của Lab 2.
4   abstract final class AppSpace {
5     static const double s2 = 2;
6     static const double s4 = 4;
7     static const double s6 = 6;
8     static const double s8 = 8;
9     static const double s10 = 10;
```

4 ViewModel lib/viewmodels/today_view_model.dart

```
47
48  final todayViewModelProvider =
49      NotifierProvider<TodayViewModel, ViewState<TodaySummary>>(
50        TodayViewModel.new,
51      );
52
53  /// ViewModel của S02. Theo dõi nốt từ đầu tháng (hoặc từ hôm qua nếu hôm nay
54  /// là mùng 1) nên hũ tự cập nhật khi có nốt mới.
55  class TodayViewModel extends Notifier<ViewState<TodaySummary>> {
56    /// Chưa có dữ liệu sau chừng này thì chuyển sang Error state (T3 → T2).
```

MVVM in the code: Service → Firebase

The data side of S02 Today. Only lib/firebase imports the Firebase SDK

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5 Service `lib/services/note_repository.dart`

```dart
/// nốt nằm trên máy với `pendingSync = true` và tự đồng bộ sau.
abstract interface class NoteRepository {
  /// Các nốt tạo từ [since] trở đi, mới nhất trước. Phát lại khi có nốt mới
  /// hoặc khi một nốt đồng bộ xong.
  Stream<List<Note>> watchSince(DateTime since);
```

6 Implementation `lib/firebase/firestore_note_repository.dart`

```dart
/// ngay trên máy, rồi tự gửi lên server khi có mạng.
class FirestoreNoteRepository implements NoteRepository {
  FirestoreNoteRepository({
    required this.currentUid,
    FirebaseFirestore? firestore,
  }) : _firestore = firestore ?? FirebaseFirestore.instance;
```

7 main.dart `lib/main.dart`

```dart
    userRepositoryProvider.overrideWithValue(FirestoreUserRepository()),
    noteRepositoryProvider.overrideWithValue(
      FirestoreNoteRepository(currentUid: uid),
    ),
    groupRepositoryProvider.overrideWithValue(
```

8 models/ `lib/models/note.dart`

```dart

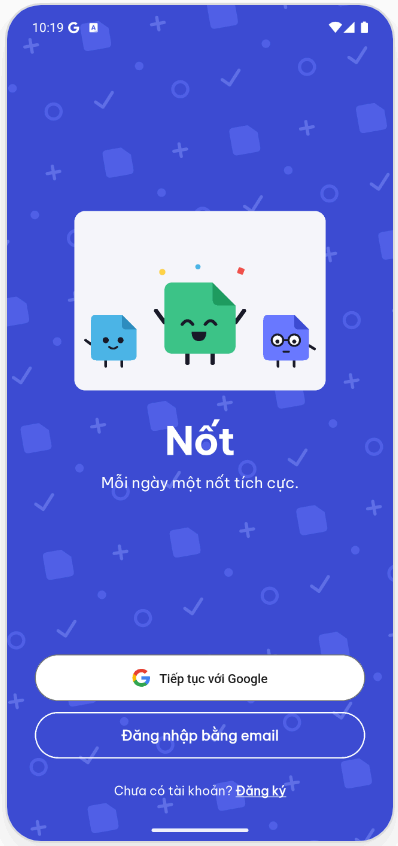
/// Một tờ nốt trong hũ, tài liệu `users/{uid}/notes/{id}`.
class Note {
  const Note({
    required this.id,
    required this.type,
```

```
$ grep -rl "package:firebase" lib/screens lib/widgets lib/viewmodels
```

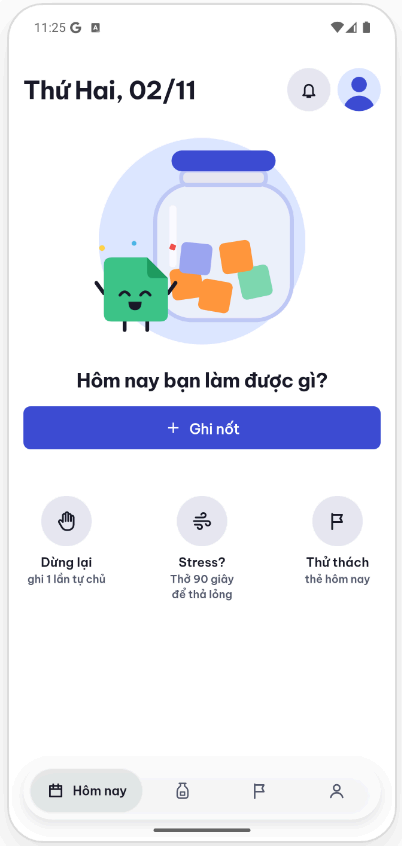
→ 0 files

Authentication

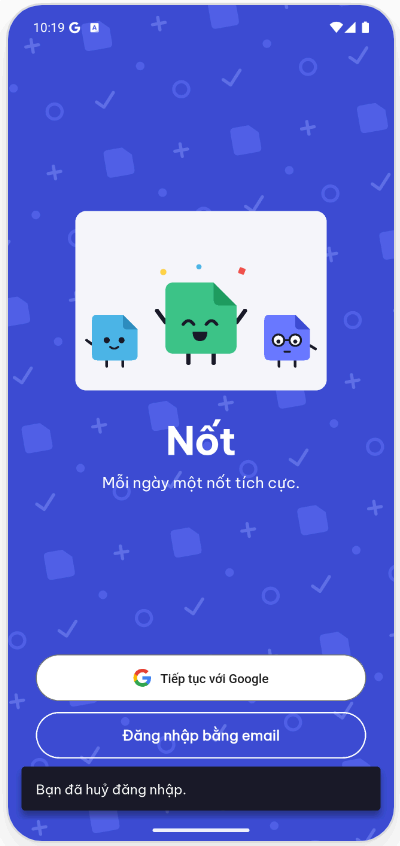
When it fails: cancel → “Bạn đã huỷ đăng nhập.” · offline → “Không có mạng. Kiểm tra Wi-Fi rồi thử lại.” · sign-out error → “Chưa đăng xuất được. Thử lại.”



Signed out → /welcome

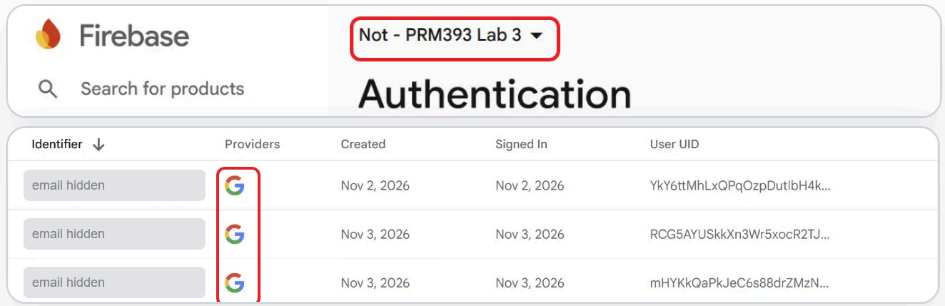


Signed in → /today



Cancelled → message

Console · Authentication › Users



Routes by auth state `lib/router/auth_redirect.dart`

```dart
final signedIn = auth.hasValue && auth.value != null;
if (!signedIn) {
  return onWelcome ? null : _withNext(AppRoutes.welcome, next);
}
if (onSplash || onWelcome) return next ?? AppRoutes.today;
```

Sign-out clears local data `auth_view_model.dart`

```dart
await _auth.signOut();
try {
  await _users.clearLocalData();
  await ref.read(draftStoreProvider).clear();
  // Hộp thư FCM nằm trên máy, không theo tài khoản: xoá để người đăng
  // nhập sau không đọc được thông báo cũ.
  await ref.read(messagingServiceProvider).clearInbox();
  ref.read(inboxRevisionProvider.notifier).bump();
```

evidence/auth/ · lib/firebase/firebase_auth_service.dart

Cloud Storage

When it fails: offline → “Chưa tải lên được: kiểm tra Wi-Fi” · signed out → “đăng nhập lại” · any other error → “Chưa tạo được thẻ tuần” + Thử lại



Failure → message + Thử lại

Console · Storage › reports/

Console file list: add after the Storage bucket is created (Blaze plan)

Rules: own folder only `firebase/storage.rules`

```
match /reports/{uid}/{file} {
  allow read: if request.auth != null && request.auth.uid == uid;
  allow write: if request.auth != null && request.auth.uid == uid
               && request.resource.size < 5 * 1024 * 1024
               && request.resource.contentType == 'application/pdf';
```

Upload to reports/{uid}, progress, URL `report_impl.dart`

```
final ref = _storage.ref('reports/$uid/$fileName');
final task = ref.putData(
  bytes,
  SettableMetadata(contentType: 'application/pdf'),
);
try {
  await for (final snapshot in task.snapshotEvents) {
    final total = snapshot.totalBytes;
    if (snapshot.state == TaskState.running && total > 0) {
      yield UploadProgress(snapshot.bytesTransferred / total);
    }
    if (snapshot.state == TaskState.success) break;
  }
  await task;
  yield UploadProgress(1, url: await ref.getDownloadURL());
```

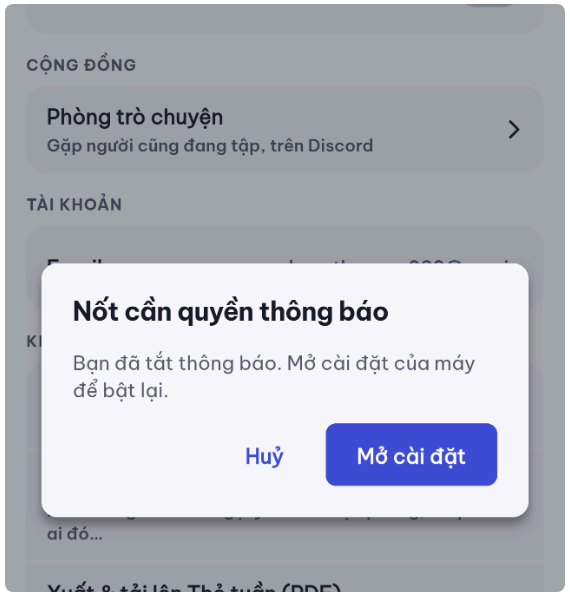
Progress on screen `weekly_card_page.dart`

```
Text(
  'Đang tải lên đám mây… $percent%',
  style: theme.textTheme.bodyMedium,
),
const SizedBox(height: AppSpace.s4),
LinearProgressIndicator(
  value: report.progress,
```

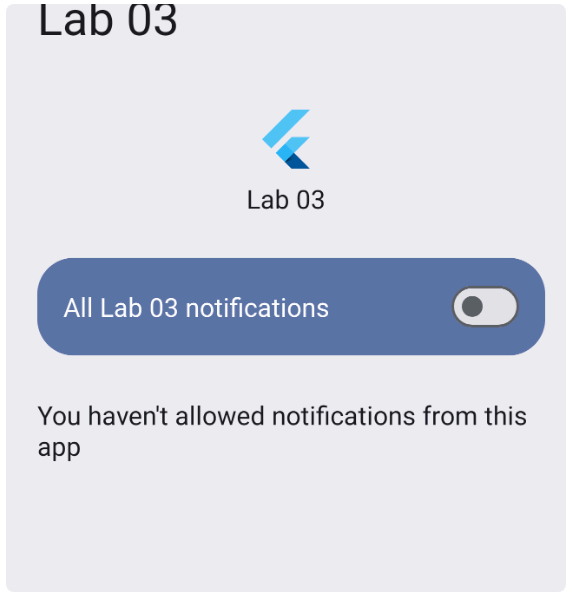

firebase/storage.rules · lib/firebase/report_impl.dart

Cloud Messaging

When it fails: permission denied → dialog + “Mở cài đặt” · unknown route → opens Today · FCM start error → logged, the app keeps running



Denied → open settings



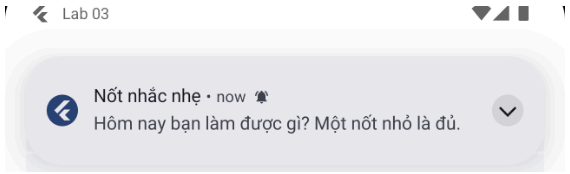
System settings



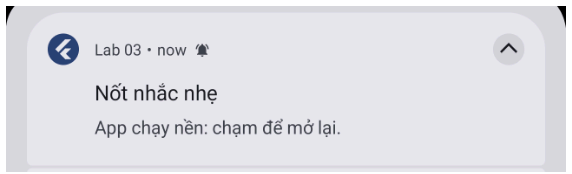
Inbox: 3 states



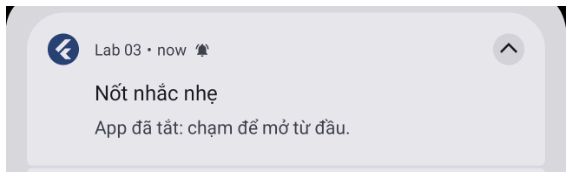
Tap · route=/jar



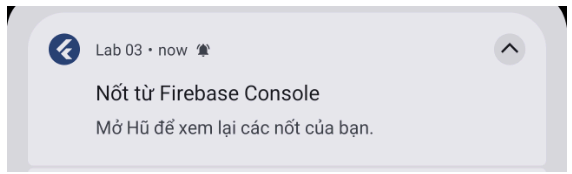
Foreground



Background



Terminated



Sent from Console

evidence/fcm/ · docs/analytics-plan.md §10

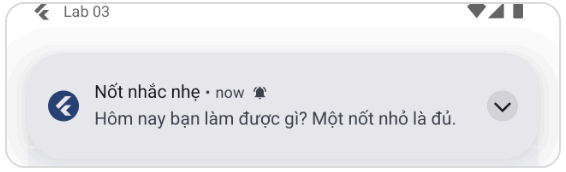
Cloud Messaging: requirements met

Each Lab 3 requirement for FCM, and the screenshot that proves it

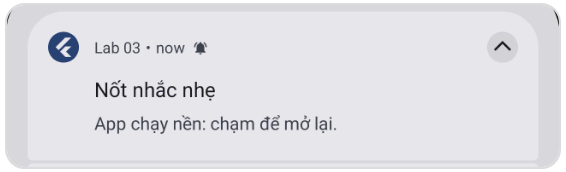
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✓ Foreground · background · terminated

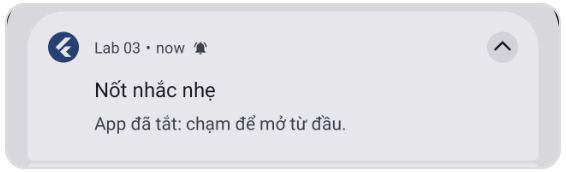
Foreground



Background

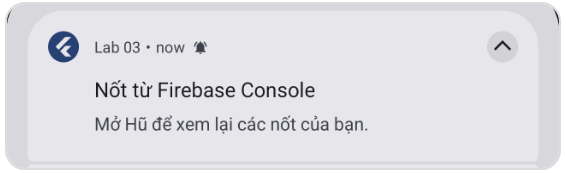


Terminated

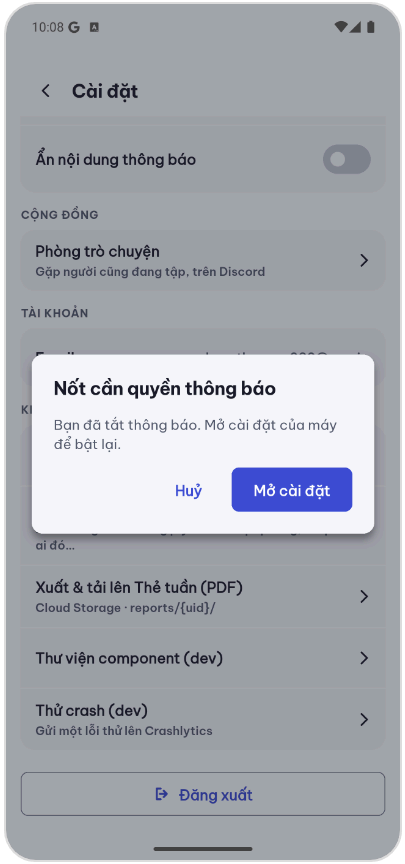


✓ Test message from the Console

Console



✓ Asks permission (Android 13+)



✓ List of received notifications



✓ Tap opens the screen

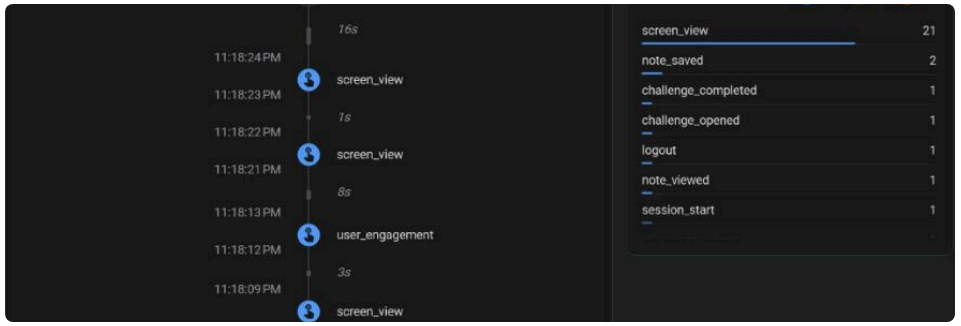


data route = /jar → Jar tab

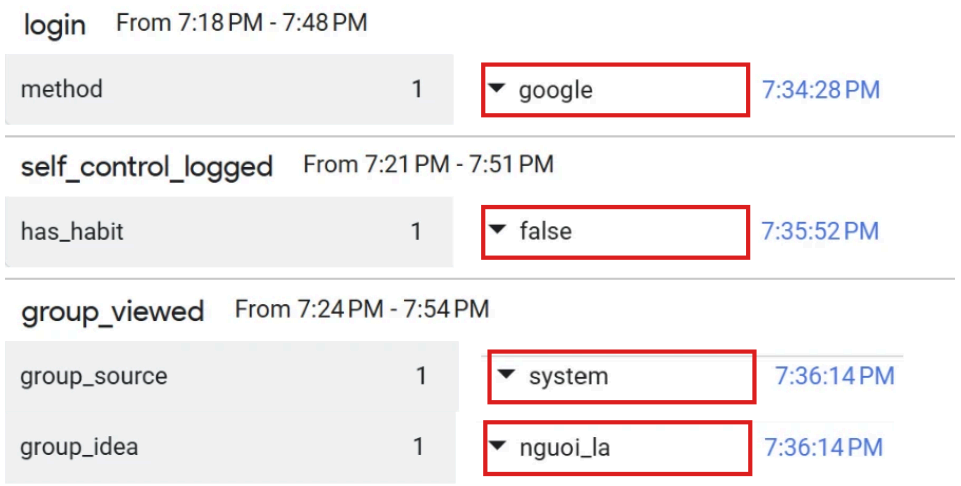
evidence/fcm/ · lib/firebase/firebase_messaging_service.dart

Analytics

When it fails: a failed log is caught and skipped (SafeAnalytics.track), so the user's action never fails



1 · DebugView · top events of one run



2 · Values sent, cut from DebugView panels

Nine custom events go through an `AnalyticsService` interface (Firebase in the app, a console logger on the Emulator). No titles, note text, emails or names are logged. The Firebase user ID is set after sign-in and cleared on sign-out.

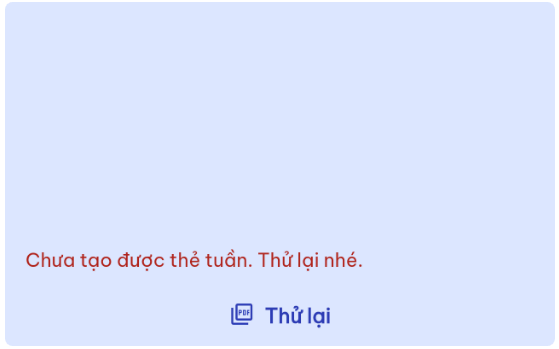
- **login / logout** · `method`
- **note_saved** · `mood, note_type, has_journal, duration_ms`. Tracks the 30-second goal: 7.7 s in this run.
- **note_viewed / group_viewed** · `note_type` · `group_source, group_idea, note_count`
- **challenge_opened / challenge_completed** · `challenge_id, note_type`
- **self_control_logged** · `has_habit`
- **report_exported** · `format`. Fires after a successful upload, so it needs the Storage bucket.

Checked three ways: DebugView (left), `adb logcat` for all events except report_exported, and unit tests that assert the parameters.

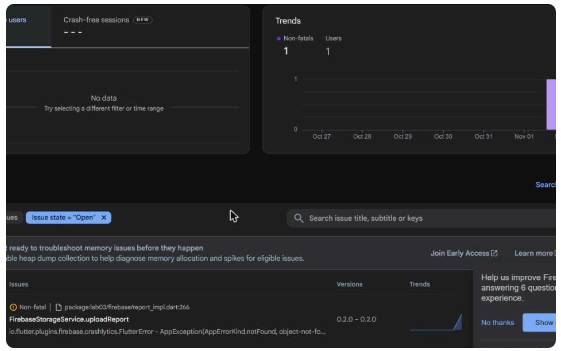
docs/analytics-plan.md §4 event table · evidence/analytics/

Crashlytics

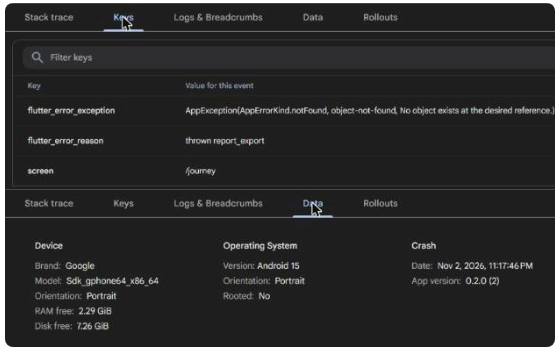
When it fails: the user still sees the error message; Emulator and web builds use a no-op reporter



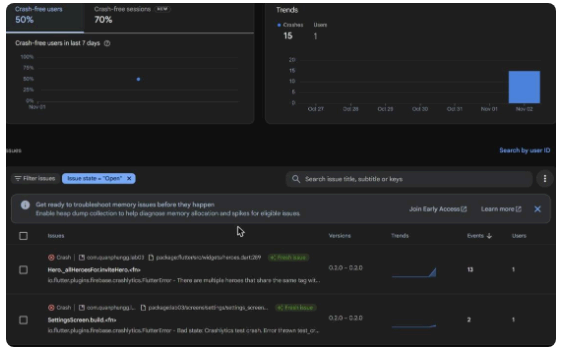
1 · User still sees a message



2 · Non-fatal in Console



3 · Key screen · user ID



4 · Test crash + a real Hero error

Crashlytics collects errors from real devices, so we see problems the Emulator never shows. Two kinds reach the dashboard:

- **Non-fatal**: the app caught the error and kept running. Exporting the weekly card fails because the Storage bucket is not provisioned; the user sees "Chưa tạo được thẻ tuần. Thử lại nhé." and the app records the error (1, 2).
- **Context on every report**: the Firebase user ID and a custom key `screen` (here `/journey`) show who hit it and where (3).
- **Crash**: a dev-only "Thử crash" button sends a test crash. The same dashboard also surfaced a real Hero error, caused by two add-note buttons sharing one tag. We fixed it with `heroTag: null` (4).

docs/analytics-plan.md §6 · evidence/crashlytics/

Remote Config

When it fails: fetch error → in-app defaults · broken JSON on the Console → default list · refresh shows “Cấu hình chưa có gì mới.”

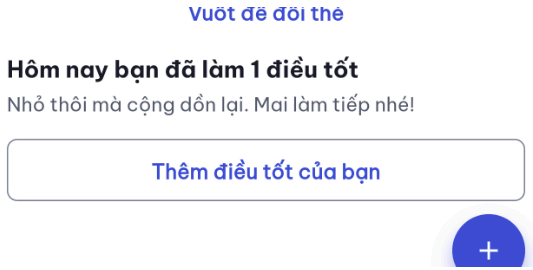
Same APK, no rebuild: only the value on the Console changed between before and after



Before · groups_enabled = true



After · groups_enabled = false

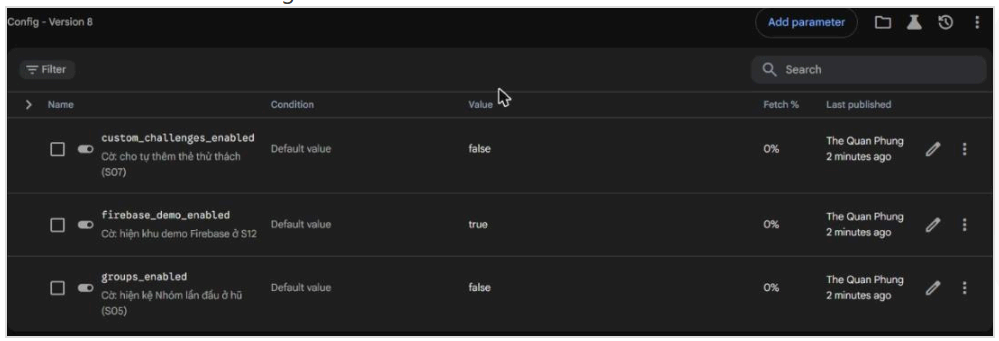


Before · custom_challenges_enabled = true



After · custom_challenges_enabled = false

Console · three feature flags



In-app defaults · used until a fetch succeeds

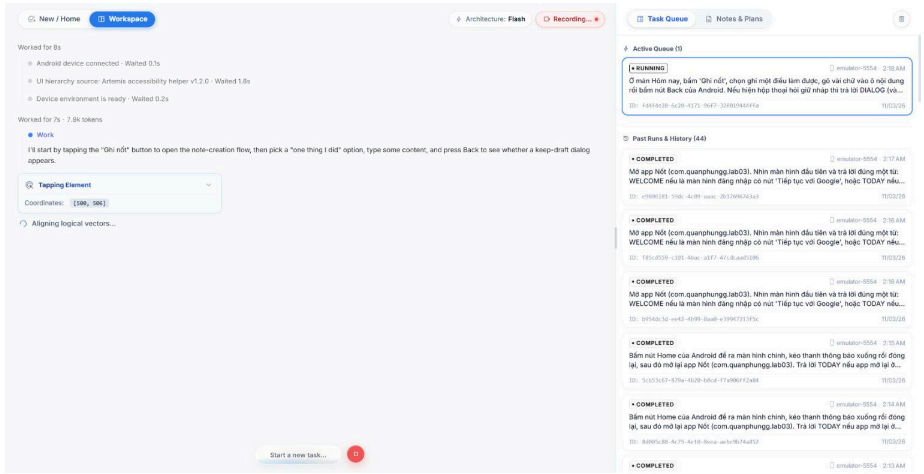
```
await config.setDefaults(<String, Object>{
  'quick_win_chips': jsonEncode(InAppRemoteConfig.defaultQuickWinChips),
  'groups_enabled': true,
  'custom_challenges_enabled': true,
  'discord_enabled': true,
  'discord_invite_url': InAppRemoteConfig.defaultDiscordInviteUrl,
  'discord_channel_map': '{}',
  'discord_greetings': jsonEncode(
    InAppRemoteConfig.defaultDiscordGreetings,
  ),
  'firebase_demo_enabled': true,
});
```

Fetch and cache · debug: every launch · release: 1 h cache

```
await config.setConfigSettings(
  RemoteConfigSettings(
    fetchTimeout: const Duration(seconds: 10),
    // Debug lấy ngay để demo "đổi trên Console, không build lại".
    minimumFetchInterval: kReleaseMode
        ? const Duration(hours: 1)
        : Duration.zero,
  ),
);
await config.fetchAndActivate();
} on Object catch (error) {
  debugPrint('[remote-config] dùng giá trị mặc định: $error');
}
```


docs/analytics-plan.md §10 · firebase/remote-config-defaults.json · evidence/remote-config/

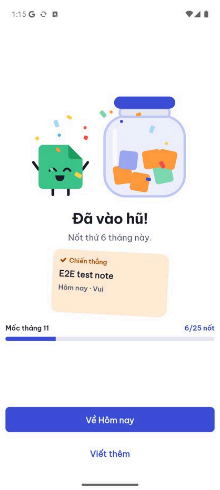
ARTEMIS E2E



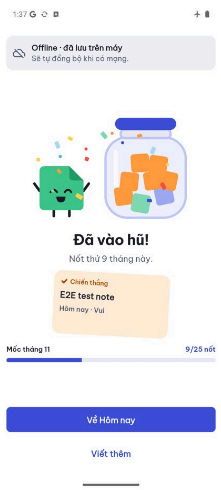
ARTEMIS Workspace · agent steps and run history (44 runs)

An AI agent (`deepseek-v4.1-flash` via OpenRouter) drives the app on the Android 15 emulator over ADB. Each case runs 3 times and asserts on the value the agent reports.

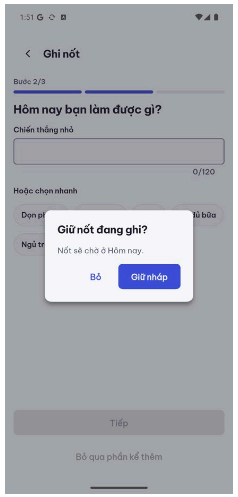
- **12 cases:** F1 save, empty, draft dialog, offline · F2 · F3 · tabs · flags · native Home and relaunch
- **35 of 36 runs pass.** T06 (offline) timed out once.
- First run was 31 of 36: two assertions in our tests were wrong (fixed, re-run 6 of 6).
- Not done yet: the 15 to 30 minute exploratory session.



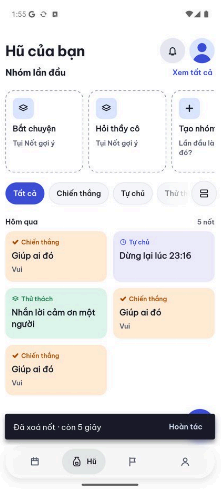
T02 · note saved



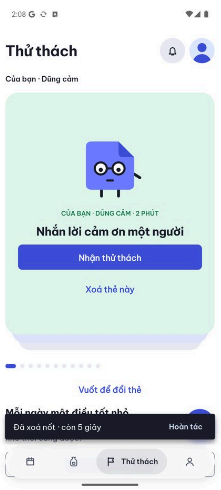
T06 · saved on device



T05 · keep-draft dialog



T03 · note deleted



T09 · challenge opened

docs/test-plan.md · e2e_test/ · evidence/artemis/

ARTEMIS: result per case

12 cases × 3 runs · 35 of 36 pass (97%) · full run 62 minutes

ID	Flow	Agent must report	Pass	Time / run
T01	Launch	`WELCOME or TODAY`	3/3	20–52 s
T02	F1 main path: save a note	`SAVED`	3/3	31–206 s
T03	Jar: open and delete the note	`DELETED`	3/3	256–391 s
T04	F1 error: empty note	`DISABLED`	3/3	55–56 s
T05	F1 error: Back with a draft	`DIALOG`	3/3 *	108–214 s
T06	F1 error: airplane mode (adb)	`OFFLINE_BANNER`	2/3	264–421 s
T07	F2 self-control	`SELF_CONTROL`	3/3	38–52 s
T08	F2 hard to stop: breathing	`BREATHING`	3/3	20–28 s
T09	F3 open a challenge, back	`DETAIL_THEN_BACK`	3/3	46–82 s
T10	Four tabs	`ALL_TABS`	3/3	33–40 s
T11	Settings: Remote Config row	`"Cờ tính năng"`	3/3	13–19 s
T12	Native: Home, shade, relaunch	`LOGGED_IN`	3/3 *	32–39 s

T06 run 3 · real failure: timed out at 420 s, no answer from the agent. Root cause not found; no sign it is an app bug.

*** T05 run 3, T12:** our assertions were wrong (the app was right). Fixed the assertion, re-ran both: 6 of 6.

Error paths: T04, T05, T06
Outside the app: T06, T12

docs/test-plan.md §5 · evidence/artemis/junit-full-run.xml · junit-rerun-t05-t12.xml

Semantics labels

41 files in lib/ set Semantics, including 17 of the 21 shared widgets

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Mood choice · button, selected, one clear label `mood_option.dart`

```dart
28 ••• return Semantics(
29       button: true,
30       selected: selected,
31       label: 'Tâm trạng ${mood.label}',
32       onTap: onTap,
33       excludeSemantics: true,
34       child: SizedBox.fromSize(
```

Today shortcuts · label and caption read together `shortcut_button.dart`

```dart
32 ••• return Semantics(
33       button: true,
34       label: caption == null ? label : '$label, $caption',
35       onTap: onTap,
36       excludeSemantics: true,
```

Offline banner · live region, announced when it appears `offline_banner.dart`

```dart
22 ••• return Semantics(
23       liveRegion: true,
```

Why

- TalkBack reads one clear name per control, not the icon and text inside it.
- `selected` and `liveRegion` tell the reader about state and new messages.
- Test tools that read the accessibility tree can find a control by its name.

lib/widgets/ · mood_option.dart · shortcut_button.dart · offline_banner.dart

ARTEMIS: written, not run yet

Five parts of the brief now have a case in e2e_test/, but no run result yet

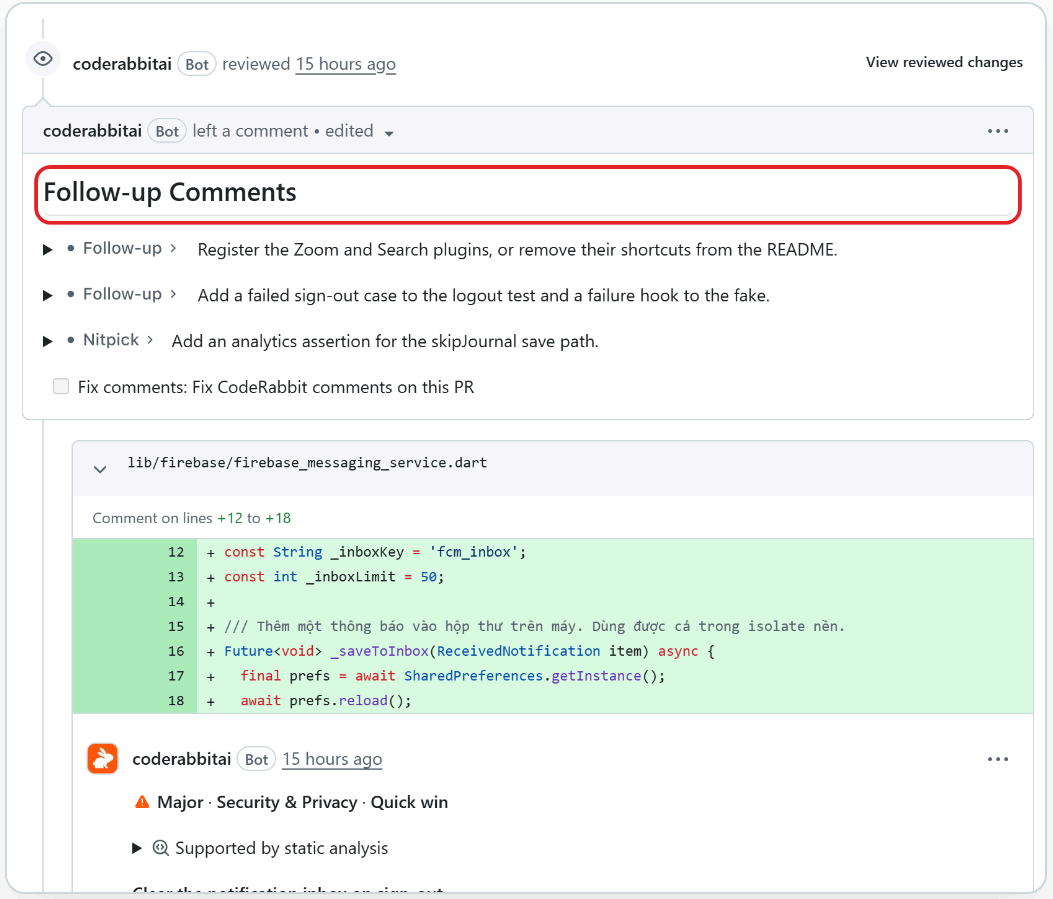
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T15 · Sign-out, sign-in	Sign out, then sign in with an email test account (not Google: its picker shows a real name and email)	Needs a test account
T16 · Wrong password	Error message under the field, then sign in with the right password	Needs a test account
T14 · Export and upload	Weekly card PDF: upload and link, or the error message + Thử lại while there is no bucket	Upload part needs Blaze
T13 · Analytics event	Save a note, then the test reads adb logcat: note_saved with mood, note_type, no note text	Needs the OpenRouter key
X01 · Exploratory	15–30 min, Pro profile; the agent explores and returns BUG lines or NO_BUGS	Not run: no bug list yet

e2e_test/test_main_flows.py · docs/test-plan.md §4, §6

AI code review

CodeRabbit reviewed every PR: 52 findings on #28, #29 and #38



1 CodeRabbit reviews every PR 52 findings on #28, #29 and #38

Góp ý của AI	Kết luận	Ghi chú
Clear the notification inbox on sign-out.	Đúng, đã sửa	736453a: `signOut` xoá `fcm_inbox` và bump `inboxRevisionProvider`; có test ở `auth_view_model_test.dart` và `firebase_messaging_service_test.dart`.

2 We judge every finding 51 right, 1 wrong · verdicts in docs/code-review.md

```
fix: clear the FCM inbox on sign-out

The inbox lives in SharedPreferences and is shared by every account on the
device, so the next user could read the previous user's notifications in
Settings. Sign-out now clears it and bumps the inbox revision so an open
sheet reloads. Addresses CodeRabbit finding 38-bfeb007a.
```

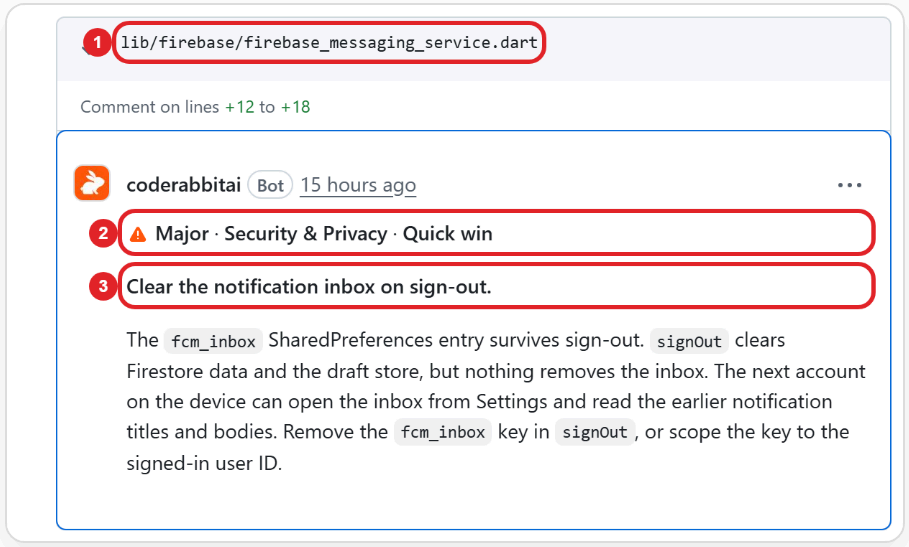
```
fix: open the right screen when a foreground notification is tapped

The heads-up shown while the app is open had no payload, and the tap
handler drops taps without one, so tapping it did nothing. Pass the route
from the message data as the payload, and let the FCM tap handler route it
too in case it is the handler the plugin keeps. Addresses CodeRabbit
finding 38-75b5e00c.
```

3 We fix with a commit 3 fixed so far; each message names its finding

AI review: what we fixed

Real CodeRabbit comments on PR #38, marked in red, and the commit that fixed each one



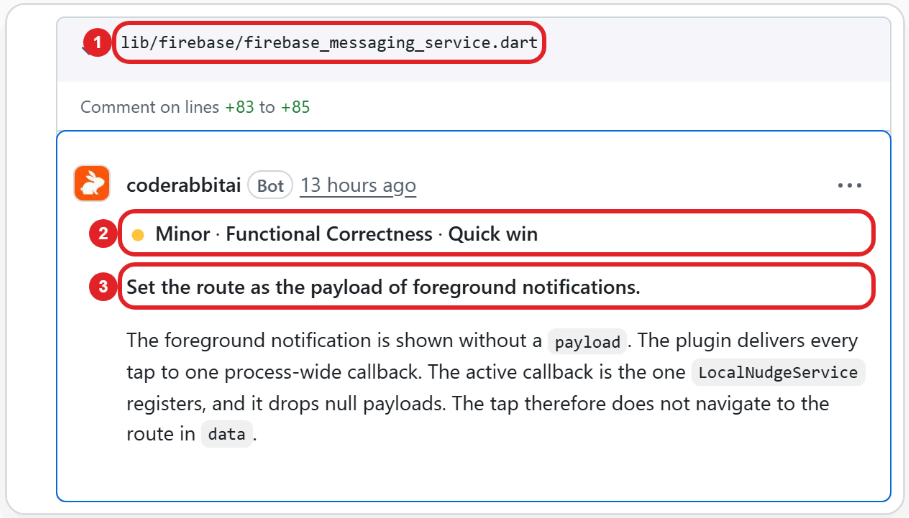
1 **WHERE**
firebase_messaging_service.dart, lines 12–18

2 **KIND**
Major · Security & Privacy

3 **FOUND**
The FCM inbox survives sign-out, so the next account can read old notifications

→

Fixed
`736453a`
Sign-out clears the inbox and the open sheet reloads. Two tests.



1 **WHERE**
firebase_messaging_service.dart, lines 83–85

2 **KIND**
Minor · Functional Correctness

3 **FOUND**
A notification shown while the app is open has no payload, so a tap opens nothing

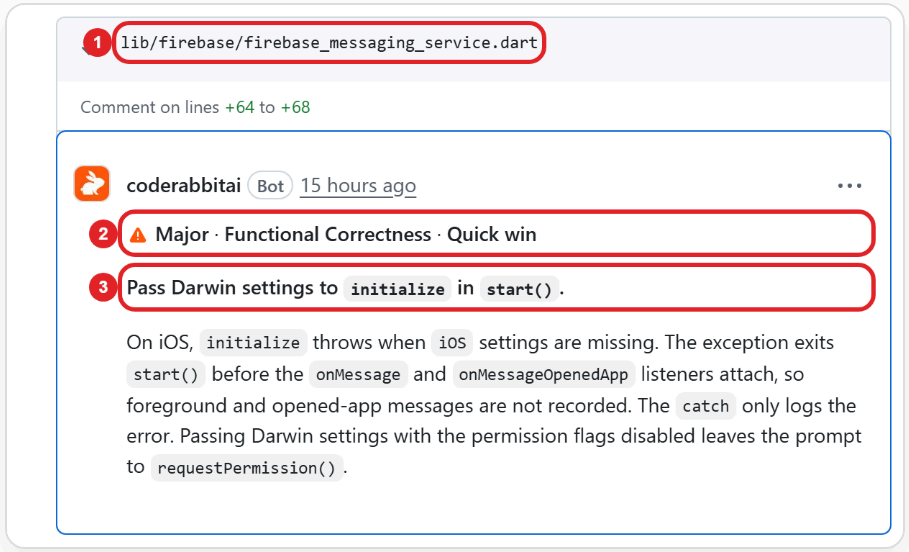
→

Fixed
`5e13651`
The payload carries the route; tapping opens that screen. Two tests.

docs/code-review.md · rows 38-bfeb007a, 38-75b5e00c

AI review: what we did not fix

Every comment was checked against the code before deciding



1 **WHERE**
firebase_messaging_service.dart, lines 64–68

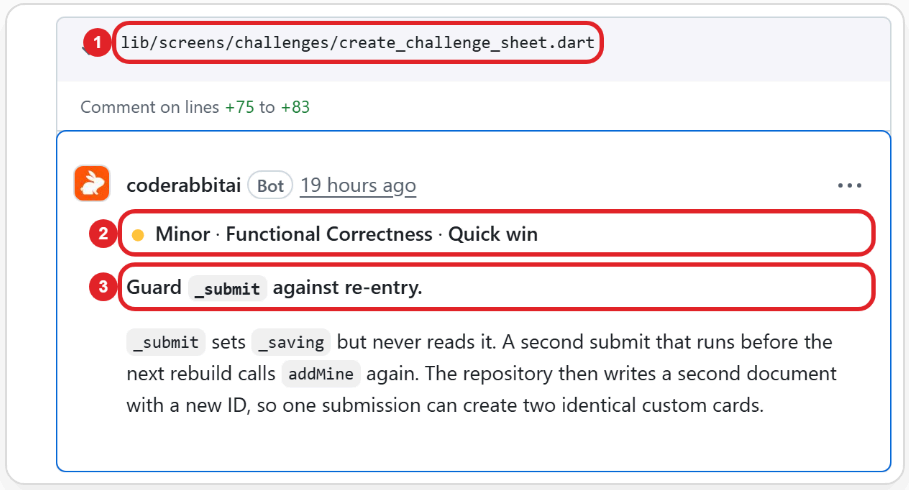
2 **KIND**
Major · Functional Correctness

3 **FOUND**
Notification setup has no iOS (Darwin) settings, so it would fail on iOS

→

Won't fix

Right for iOS, but the app ships on Android and web only. Major is too strong here.



1 **WHERE**
create_challenge_sheet.dart, lines 75–83

2 **KIND**
Minor · Functional Correctness

3 **FOUND**
_submit sets _saving but never reads it, so one tap could add two cards

→

Rejected

The AI is wrong: canSubmit = !_saving && … already disables the button while saving.

docs/code-review.md · rows 38-cfffb9a5, 29-c0bdd08c

Lessons learned

1

Test on the real project early

Emulator has no Analytics or FCM

2

Design docs can disagree

Pick one, log it · D-11

3

Rename what users misread

"Bỏ qua phần kể thêm" · D-28

4

Check the AI tester too

4 of 5 failures: our own asserts

5

Read AI review before merging

1 of 40 bugs fixed at merge

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Thank you!

AI tools used

Claude Code code, docs, slides

CodeRabbit PR review

ARTEMIS E2E agent

DeepSeek ARTEMIS model